48,000 pounds, and the resistance for both angles is 96,000 pounds.

The true value for the moment of inertia in this case is 16.59, and the value for y is 3.92 inches. Therefore, the true value for the bending moment $\frac{fI}{y}$ is $\frac{16{,}000 \times 16.59}{3.92} = 68{,}000$ pounds, and the resistance of both angles is 136,000 pounds.

A Reinforcing Flue Sheet.

A reinforcing flue sheet for a locomotive boiler has recently been designed and placed in service with good results by Mr. J. A. Doarnberger, master boiler maker, Norfolk & Western Railway, at Roanoke, Va. The device consists essentially of a second flue sheet placed about 8 inches in front of the back flue sheet, of somewhat thicker material than the latter, and thoroughly fastened to it with stay-bolts. The flues extend from the front tube sheet to the reinforcing sheet, being simply expanded at both ends without beading. A short section, or detachable safe end, one end of which is swaged down to fit inside the main part of the flue, connects the reinforcing sheet with the back flue sheet.

This construction accomplishes several desirable results. In the first place, the reinforcing sheet forms a strong support for the back flue sheet, so that it is practically impossible for it to bulge. It provides a detachable safe end for the flues, so that when a flue wears out at the back flue sheet, as it commonly does, it is not necessary to remove the entire flue and reweld a new safe end onto it, but the short detachable safe end can simply be cut out and another substituted in a very short time. Furthermore, there is by no means as much strain on the flues, due to expansion and contraction, tending to cause leakage at the fire-box end. In fact, it is simply necessary to roll the flues at the back flue sheet without beading them over, as they simply have to withstand the collapsing pressure due to the steam pressure in the boiler, and not any longitudinal stresses due to the expansion or contraction of the tube. For this reason the flues can be used even after the beads, or the ends which are allowed to project an eighth of an inch or so through the flue sheet in case the flues are not beaded, are entirely burned away before it is necessary to replace the safe end.

Another advantage claimed for the reinforcing sheet is that it provides a means for better circulation of the water around the back flue sheet, since it forms a narrow channel through which the water will tend to flow.

The outline of the reinforcing sheet is shown by the dotted lines in Fig. 3. It is evident that considerable care and skill must be used in drilling the tube holes in this sheet, so that they will correspond exactly with the holes in the back flue sheet; and in order to accomplish this, the reinforcing sheet is usually bolted to the back flue sheet and the holes drilled coincidentally.

The flues in all the locomotives which have so far been equipped with this reinforcing sheet have been spaced in vertical rows. In the case of the boiler shown in the illustrations there are two hundred and fifty-eight 2¼-inch flues, spaced 3⅛ inches between centers. Since these are placed in vertical rows the bridges are ¾ inch wide. In the center of every rectangle, formed by four adjacent flues, a stay-bolt is placed. These bolts, therefore, are also in vertical rows, 3⅛ inches apart, and their position does not decrease the width of the bridges. These stay-bolts are ⅞-inch diameter at the center, and are threaded at either end and riveted over both flue sheets.

In the earlier applications of the reinforcing sheet ordinary flues were used, extending from the front tube sheet to the back tube sheet, and these were expanded into the reinforcing sheet. This provided ample support for the back flue sheet, as was intended; but, of course, the expense of the flue work was in no way decreased. The idea was then conceived of cutting off the main flue at the reinforcing sheet and using a short section swaged down at one end to form a telescopic joint at the reinforcing sheet, forming a detachable safe end. In the latest design of this construction Mr. Doarnberger has used a heavier auxiliary sheet and has butted the flues and the safe end. Expanding a flue which extends only part way into the flue sheet in this way seems to give a joint with sufficient strength, as in one case the end of the flue which was ex-

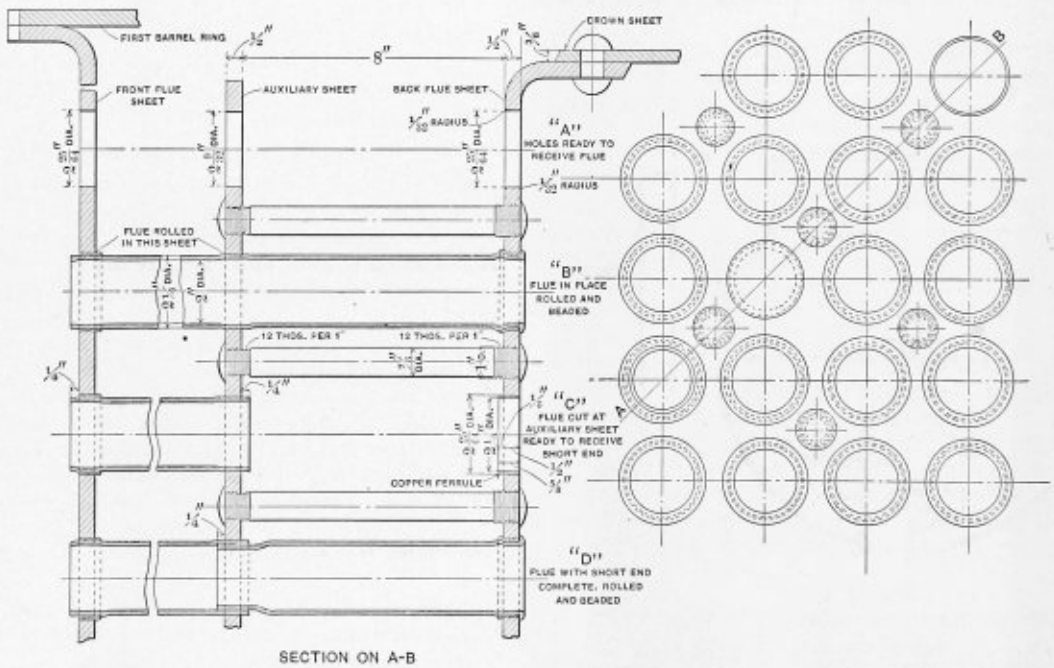


FIG. 1.—REINFORCING SHEET WITH TELESCOPIC JOINT BETWEEN FLUE AND DETACHABLE SAFE END.

use of the roller and prosser expanders. The work is usually done as follows:

First, prepare the flue hole by removing the scale or burr from flue hole; also true up the hole by using a reamer made for that purpose. Then remove the sharp edge from the flue hole inside and outside of back flue sheet. Put in copper ferrules, allowing them to project 1/32 inch inside of face of flue sheet. Expand ferrule into flue hole, using sectional expander.

The flues are prepared and swedged to fit into the copper ferrules nicely by applying three or four blows with a driving bar, made for that purpose. They are opened, or mandreled out, as you prefer to call it, with a sectional expander with the hub of expander ground off and used as mandrel. Flues are allowed to extend 3/16 inch for bead.

The flues are rolled with the use of air motors, and flaired over by the use of what we call a bob, made expressly for that purpose, used in connection with a long stroke hammer. They are then prossered by the use of a jam riveter; the prosser turned three times in operation, and are then calked by standard beading tool, made to gage, used with pneumatic hammer.

This report recommended that whenever flues leaked, the leaks should be stopped by the use of the sectional expander, and that when the tubes are poor or thin, and, in the judgment of the workman, there would be a possibility of bursting the tubes, the sectional expander should be used with the shoulder ground off. In case the tubes are so poor that leaks cannot be stopped with the sectional expander they are removed. The assertion was made that flues cannot be put in locomotive boilers better than they are at the present time, and the fact was pointed out that with an iron shim wrapped around the flue for a ferrule, and with the flue only rolled, not beaded, the flues in the front end of the boiler give hardly any trouble at all. The cause for the leaking of flues at the fire-box end is due to a number of things, principally to bad water and great variation in temperature. It was recommended that when a fire is cared for it should be placed forward and the front grates kept covered to prevent cold air coming in contact with the flues and back flue sheet.

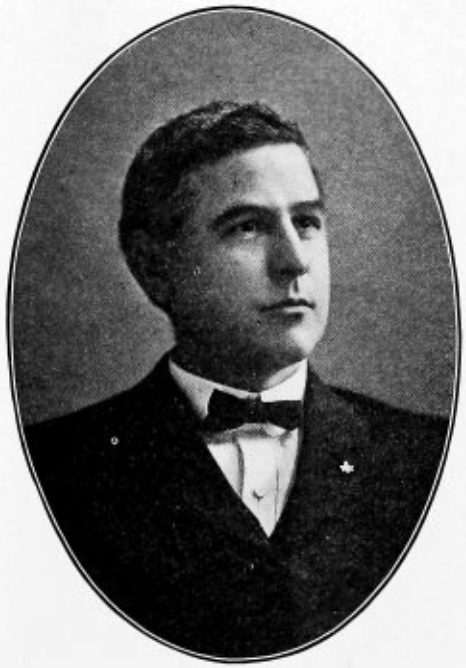
The report of the committee was accepted and the subject was opened for discussion.

Mr. A. N. Lucas—We use a prosser expander exclusively in the fire-box. We do not use the roller at all in setting new flues or repairing old flues. I think a great deal of damage in the roundhouse to the flue and to the bead is due to heavy expanding. After you have set the flue properly at the first setting with a prosser expander, it has the full contour of the expander, and when the flues leak it certainly does not require the heavy pounding used at first. I think if the flues in the roundhouse are cared for by using a lighter maul, the same hammer you use for pounding up large stay-bolts, turning your expander two or three times and then beading lightly you will get very much better results than if you go after them and "hit 'em hard."

Again, when you expand your flue you use a 5/8-inch expander, that is, from shoulder to ball; the bead is not down against your sheet, and it expands the flue inside of the sheet properly. Now if you use that same expander on repair work it will go inside of the flue 1/16 or 3/32 inch farther than it did when you first expanded the flue. That will tend to ball the flue up farther inside, but it doesn't fit in the original groove, and consequently you do not get the results you are looking for. I believe we should use a shorter expander in the roundhouse after the flues are beaded.

The President—There is one thing in regard to this subject I would like to refer to and see if the members have an opinion in regard to it, and that is the exclusive use of a sectional expander. It is a very important question in the East, and the opinions of the members of this association will be valued by a good many people who handle flues in the Eastern districts.

Mr. Roesch—We are just as interested in preventing the flues from leaking as in doctoring them after they begin to leak. Now we have been frequently criticised for poor workmanship, when probably the entire trouble was poor handling of the boiler after it is in service. Have you ever noticed the intimate relationship that exists between washing a boiler and flue leakage? We speak of bad water causing flues to leak. That is true, because the worse the water is, the more frequently we have got to wash the boilers, consequently the shorter the life of the flues. I have been pretty much interested in the flue problem practically all my life, because at one time I was connected with a railroad that, on one certain division, had no flue troubles whatever. We put the flues in and forgot about them. On another division of the same



A. E. BROWN, FIRST VICE-PRESIDENT.

railroad, if a set of flues passed three months we did pretty well. I noticed on the division where we had no flue troubles that we never washed a boiler. Now that is a strange proposition, but it was up in the mountains, with absolutely pure snow water, consequently there was no mud in the boiler. On the other division we had an alkali water which made it necessary to wash the boilers every trip.

Mr. M. E. Wells, who is an authority on locomotive boilers, made the statement at one time that the leaking of flues was all due to the difference in the temperature of the boiler; that is, the difference in the temperature of the water itself. When I became connected with the Southern Railway I thought I would see what that difference amounted to. I made some trams so that I could tram the length of the flue immediately after the fire was knocked, and at the same time tram the length of the boiler between the flue sheets. I found that shortly after the fire was knocked, the flue contracted more than the boiler; that is, it pulled in the boiler, probably 1/64 or 1/128 inch, because the cold air passing straight through the flue contracted it while the boiler was still full of hot water, and was protected by a jacket and lagging on the outside, which kept the boiler from cooling as quickly. Finally, however, the boiler commenced to contract and contracted more than the flue; in other words, it started to push the flue out of the flue sheet. I found the difference in the

contraction to be about 1/64 inch in the first contraction. That is, the flue shortened about 1/64 more than the boiler, then the boiler shortened more than the flue, then finally the flue shortened, after the water became cold, until they both became the same length again. Now that shows that there is this pushing and pulling action going on in the boiler all the time.

On account of the back flue sheet being exposed to a greater degree of heat, we have all our troubles in the rear end. I thought I would see what effect the stopping of the washing of boilers would have on this. We have good water, only mud in it, no scale, no encrusting material or anything of that kind. I put a lot of draw-off cocks on the engines, and engines that we used to wash out every week ran thirty days, some of them sixty days. We never knock the fires any more, and we don't have to send the boiler makers out any more. Therefore it seems that there is an intimate relationship between the number of times the boiler is washed



J. J. FLETCHER, SECOND VICE-PRESIDENT.

and the number of times the flues leak, or the number of times that you have to renew flues, and of course the worse the water is the shorter the life of the flues.

Mr. Flavin—I find that in many places they are overcoming a great deal of flue trouble by using the waste water from the boiler for washing out; in fact it has become a standard practice throughout the entire West. There is no question about the expansion of the flue sheet; it must be accommodated. It was brought to my mind very forcibly in a condition wherein we received some engines furnished with 5/8 flue sheets. In every case in which I renewed these flue sheets I used a 1/2-inch flue sheet, and in smaller engines 7/16, merely to accommodate that necessary expansion, and I found that with flues 19 or 20 feet long, instead of the sag having a tendency to loosen the flues, I got better service out of the long flues than the short ones.

I believe that the action of the prosser expander can be governed entirely by having a shoulder resting on the flue sheet proper, as well as on the bead of the flues. In that way the prosser expander will always enter the flue the same distance, and we will always make the same joint unless the sheet is worn to such an extent as to make a vast difference. As to the use of the prosser expander alone, while this is of course endorsed in boiler practice throughout the country where there is ordinarily bad water, or fairly good water, it probably is successful in the case of the first one or two or three sets of flues, but where conditions exist where a flue sheet will receive seven or eight sets of flues before it is removed, the holes gradually become oblong, and there is no prosser expander on the market to-day that does not expand centrally, where the roller will roll irregularly and therefore accommodate itself to the oblong hole. For that reason I believe after a flue sheet has received three or four sets of flues that the use of the roller, even lightly, will insure better work than the use of the prosser expander alone, although in all other cases I would say the use of the prosser expander is best.

Mr. A. N. Lucas—Just a few words, to show how much a flue expands. We were unfortunate in the roundhouse foreman having his man fire up two or three engines without any water in them, and upon inspection I found four or five flues with the bead sticking right out straight, while with the balance of the flues the bead was up against the sheet. On close inspection I found these flues were stopped up and the heat didn't get through them, the balance of the heat going through the rest of the flues pulled this bead right out straight, and when the boiler cooled off the bead was still out.

Mr. Bennett—The Boiler Manufacturers' Association, in their rules covering flues, have one provision stating that a flue should be rolled tight and no more. I believe the tendency of applying flues with air is to overwork them. When a man is doing it by hand he knows pretty nearly when a flue is rolled enough or expanded enough. Now when he uses an air machine he does not realize how much rolling the flue is getting. The same is true when he prossers the flue with a long-stroke hammer; the hammer is doing the work, not the man. After a flue has been prossered just so tight and properly beaded, all further efforts along that line will only wear the flue out. I think in a good many cases there is so much work done on the flues when they are applied, that it takes about all the substance out of the flue and nothing is left for the roundhouse men to do. When rolling a flue with air, you will sometimes find a crease 1/32 inch on a side; that means a reduction in the thickness of the flue of 1/32 inch.

What Mr. Lucas says about overworking the flue is very true. You hammer the prosser in there every time flues are leaking—it isn't necessary. All you can do is just to get the flue to the sheet. Further efforts along that line only go to break the unions and destroy your flue sheet.

The idea that this committee has reported about flues leaking on the back end and not on the front end is very true. Just as soon as a flue leaks on the back end, it is up to the boiler maker; or, in other words, it is considered more work. In the front end, as the committee says, all you have got to do is to put an apprentice boy there, if necessary, and roll the flues, and you will never have any trouble with them leaking. It goes to show that heat and the change of temperature in the back end cause them to leak.

I think we are applying flues to-day just as well as they can be applied. The practice on the New York Central is to use the prosser exclusively in roundhouse work, and we have latterly adopted as standard 1 7/8-inch flue holes and use the 15 gage copper and the flue is swedged down; that would give us 1 3/4 inches in the back end, and the flue is swedged straight for 1 inch, and from 1 inch to 4 inches is tapered; it is a tapering swedge.

Mr. Conrath—Flues should be prossered only when applied. When flues have been prossered with a Missouri Pacific standard prosser, as we call it, with a tapered shank on the back to turn the end of the flue to a 45-degree angle, it is then turned the rest of the way with a beading tool and finished. We next apply the pressure, 25 pounds in excess of steam pressure carried. Any flue that shows a leak is lightly rolled. When flues are being applied we use a copper 3/8 inch wider than the sheet, the copper extending into the water side, while the prosser work will close the end of it tightly around the flue so that it is watertight. It is easier to make a joint with

the prosser against the sheet by bending the copper over against the inner part of the sheet. Also the sheet should be lightly counter-bored so as to take off the burr on the inner side of the sheet. Then the flues are worked into the center.

We always mark a diamond in the center of our flue sheets. They are obliged to work the flues around that diamond shape first, and then in the center last. You will all admit that no matter how large the stretch, the expansion tool which you are working the flues with will stretch that sheet. Consequently the men will start on one end of the flue sheet and work the flues completely into the other end, the opposite corner. The result is that the expansion has worked into that corner and you have got oblong holes just as soon as your sheet becomes hot and your pressure is on there.

The cold blast every time the fire door is opened will strike the bottom flues and cause them to leak quicker than the others. You begin and roll your flues on the bottom when the engine is warm. You are stretching your sheet every time you roll it. You are not only stretching the sheet but spreading out and enlarging your holes. After that has been done for a dozen times the holes right above that space become oblong. They are not giving any trouble up there; the beading tool will supply it for the time being; eventually the beading tool will not do the business; the boiler maker gets his roller, sticks it in the flue and the tool will not turn because the flue is oblong. The next thing, away goes the pin and away goes the bridge. That should be avoided, not using the rollers any more than necessary.

My method of working flues after they have been out on the road is this: we first use a ¾ taper mandrel, ¾ to the foot; the same taper the roller pin would be. The holes are all round as they come from the shop. The pin can be used for a little while; then use the prosser. Eventually your flue beads have become spongy; you have got to use the roller; then use it lightly or with good judgment; and after you have done that you have got all out of a set of flues that you will get out of them. I have tried the rollers alone, I have tried the prosser alone, and I found that by using the pin somewhat, the prosser and the rollers last and the standard beading tool, that we get better results than any other way of working flues.

We find when we are applying the flues that we have got perhaps half a dozen different size holes in the front sheet. In order to make a standard swedge, maintain that swedge and the standard tools, we fill these holes out with different size coppers; we start with a No. 14 gage down to a No. 10, then enlarging the copper as the hole becomes bigger, and maintaining the same size swedge, which is 1⅞. We generally allow 3/16 inch strong for bead. If necessary, on the bottom we sometimes allow a little bit more in order to brace the sheet.

The thickness of copper is No. 14, which is 40 pounds, from that down heavier. No 14, 13, 12, 11 and 10. No. 10 is a 70-pound copper. They differ 1/64 inch every number. Where there are different size holes we fill out the holes with different size copper. We always maintain the same swedge.

There is one more thing in regard to leaky flues. All the boilers in the land won't keep flues tight if you do not keep the nozzles and steam pipes tight. Trouble with the front end makes the flue trouble at the back end. Follow up leaky steam pipes and watch them closely.

Mr. D. A. Lucas—My experience has been that the best results are obtained with the prosser expander. The roller ought to be left to the judgment of the man in charge. Referring to the recommendation of the committee, there is one thing in particular to which I would seriously object, and that is grinding the boss off of the expander. I think that is doing away with all mechanical part of working the flues, making a tool that is entirely out of mechanical lines.

I am getting good results out of flues with a wide copper, which I would recommend by all means, and I would also recommend not lighter than a 40-pound copper, something that will stand the tension on the joint, put inside of the sheet.

Mr. Filcer—There have been some remarks made here relative to flues stopping up. Some months ago we adopted 1⅞-inch holes, in the place of 2 inches, in order to increase our bridges and get a little more water around the flues, and since that time we find we have no stopped-up flues, and we also find we have less leaks with the 1⅞-inch hole than we did with the other.

Mr. German—A little over a year ago, on the Lake Erie & Western, they were having considerable trouble with their flues leaking. As it is a bad water district, I asked the privilege of putting in a flue in a certain way. The bridges that were in the sheets at that time were only ⅝ inch. Reducing



A. N. LUCAS, THIRD VICE-PRESIDENT.

the flue from 2 inches down to 1⅞, gave a little better water circulation there and we kept the rolls out of the flues entirely. We didn't even use a pin to fasten them in the sheet. Instead of using a pin we used a straight expander to fasten the flues in the sheet, then we would bell them out with a flaring tool, then put in the recess prosser and prosser them, turn it a few times, that would turn the flue over sufficiently to put the beading tool on and calk it. By that method, leaving the flue below its original size in the application, we increased the mileage on that road from ten to thirty thousand miles.

In regard to cleaning the flues, this flue acts almost the same as a reduced nozzle; it creates a kind of a vacuum that just simply cleans the flue out. We don't have one, I will say, where we formerly had ten flues stop up since that system was adopted. This has been adopted now nearly all over the New York Central lines. I advise using the prosser from the beginning to the end. Since we have adopted the sectional expander, we do not have one roll out of twenty that will hold in any sheet.

Mr. Brown—The swedge which Mr. Bennett refers to, I feel is practical, so far as a swedge is concerned, but I would like to ask why we can sit here and advocate a copper ferrule ¾ or ⅞ inch wide? What use have we for it? You have